Title: Quantum Networks Meets Quantum Random Number Generators: A Framework for Quantum-resistant TLS Deployment

Year: 2022

Authors: S. Fernandez, R. Vasquez (School of Information Sciences, Lakeshore University)

Keywords: quantum networks, cryptanalysis, post-quantum cryptography, quantum key distribution, BB84

Abstract:

Quantum-resistant TLS deployment remains a critical challenge for quantum networks, particularly in banking infrastructure. This paper proposes a framework built on quantum random number generators that explicitly accounts for side-channel attacks on post-quantum implementations. We evaluate the framework on a simulated 100 km fiber QKD channel and compare it against 7 baseline methods. Experimental results show that the proposed approach improves logical qubit count by 23.7% while keeping secret key rate within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of quantum networks has changed how researchers and practitioners approach quantum-resistant TLS deployment, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, quantum random number generators have attracted sustained attention as a promising route toward solving quantum-resistant TLS deployment in realistic settings such as banking infrastructure. Despite considerable progress, existing solutions for quantum-resistant TLS deployment tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in signature size frequently come at the expense of decapsulation time, forcing designers to accept trade-offs that are poorly understood.

This paper addresses these gaps by studying how quantum random number generators interact with isogeny-based protocols when applied to quantum-resistant TLS deployment. Our work is motivated by the growing demand for banking infrastructure, where reliability and transparency requirements are especially strict. Understanding side-channel attacks on post-quantum implementations is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

OConnor et al. (2018) introduced an early framework based on quantum random number generators, reporting a 38.4% improvement in secret key rate over previous baselines. A survey by Kowalski et al. (2012) categorised existing approaches to quantum-resistant TLS deployment into three families and highlighted the lack of standard benchmarks. Researchers working on quantum networks have explored isogeny-based protocols as an alternative, with Mukherjee et al. (2016) noting that the approach scales poorly beyond 32 components. Kowalski et al. (2010) analysed side-channel attacks on post-quantum implementations and showed that purely heuristic solutions cannot guarantee consistent logical qubit count. More recently, Tanaka et al. (2015) combined quantum random number generators with isogeny-based protocols, achieving a 17.0% reduction in overhead on a simulated 100 km fiber QKD channel.

Several studies have criticised the reliance on synthetic workloads, arguing that results on a simulated 100 km fiber QKD channel may not generalise to banking infrastructure. In contrast to these efforts, our work treats quantum-resistant TLS deployment and side-channel attacks on post-quantum implementations jointly rather than as independent concerns. Industry reports corroborate these academic findings, pointing to

quantum-resistant TLS deployment as a persistent obstacle for banking infrastructure. Despite this body of literature, no prior study has systematically compared quantum random number generators and isogeny-based protocols under identical experimental conditions.

3. Methodology

We formalise quantum-resistant TLS deployment as a constrained optimisation task in which the objective is to maximise logical qubit count while keeping logical qubit count within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on quantum random number generators, and a validation layer that incorporates isogeny-based protocols. All experiments were conducted on a simulated 100 km fiber QKD channel, which provides 16 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 10 times with independent random seeds, and we report mean values together with 95% confidence intervals.

Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data. The core routine applies quantum random number generators to reduce the effect of side-channel attacks on post-quantum implementations, followed by a calibration step that stabilises secret key rate. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 7 baseline methods, including a conventional approach built on isogeny-based protocols and two recent variants reported in the literature.

The proposed design improves logical qubit count by 23.7% relative to the strongest baseline and reduces secret key rate by 10.1%, a statistically significant difference ($p < 0.05$). Ablation experiments show that removing the isogeny-based protocols component reduces logical qubit count by approximately 14.1%, confirming its contribution. A sensitivity analysis over 128 parameter settings shows that performance remains stable, with variation in logical qubit count below 12.9%.

4. Conclusion

This paper investigated quantum-resistant TLS deployment through the lens of quantum random number generators, and demonstrated that a carefully designed pipeline can improve logical qubit count by 23.7% on a simulated 100 km fiber QKD channel. The findings suggest that quantum networks will continue to benefit from closer integration of quantum random number generators and isogeny-based protocols, particularly in banking infrastructure. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of side-channel attacks on post-quantum implementations. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for quantum-resistant TLS deployment.